

RESEARCH ARTICLE

Smart Tourism Recommender System Modeling Based on Hybrid Technique and Content Boosted Collaborative Filtering

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ABSTRACT A recommender system (RecSys) is a smart solution that offers personalized items to users. Utilizing adequate information from historical transactions in the tourism industry such as items, users, ratings, and reviews becomes valuable input in providing personalized RecSys regarding smart decision-making for users in tourism. This study proposes a new approach for tourism RecSys development through a hybrid model combining User-Based Collaborative Filtering (UBCF), Demographic Filtering (DF), Aspect-Based Sentiment Analysis (ABSA), and Content-Boosted Collaborative Filtering (CBCF). This model utilizes six steps in the Cross-Industry Standard Process for Data Mining (CRISP-DM): business understanding, data understanding, data preparation, modeling, evaluation, and deployment. A tourism dataset has been produced by combining raw data from the TripAdvisor website with demographic information from Google Maps to enhance user profiles. The model performance was evaluated using the measurement of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) through the 5 cycles of 10-Fold Cross Validation. Based on the average results of MAE and RMSE, CBCF has achieved performance improvements respectively at 84.7% and 82.3% compared to the performance of UBCF using 100% dense UI-Matrix. This study has succeeded in proposing a novelty model for tourism recommender systems in avoiding the cold-start problem and sparse matrix reduction through the synthetic data generation regarding performance improvement of the recommender system complemented with related aspects of recommended attractions.

INDEX TERMS Collaborative filtering, data mining, demography, recommender system, sentiment analysis, tourism.

I. INTRODUCTION

A. TOURISM INDUSTRY

The tourism industry has significantly contributed to economic growth in various countries, especially before the COVID-19 pandemic [1]. A large part of this economic growth in many countries comes from the tourism sector [2]. The arrivals growth rate of international visitors was reported by UNWTO to be around 4% in 2019, accounting for approximately 60 million arrivals [1]. In the post-pandemic era of COVID-19, the tourism industry has started to rebound as

predicted in a previous study [3]. Various strategies aimed at improving service quality for tourists are being pursued to enable the tourism industry to produce a positive impact on regions. As an effort to improve the quality of service, this can be provided by improving the tourist experience by utilizing smart technology [4].

B. TOURISM EXPERIENCE

Obtaining tourism product information is important for tourists before deciding to make a tourism visit. Various conventional approaches are taken to obtain this information, either through information available in brochures, on

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websites, asking directly to travel agents or other people they know. For tourists, an effective decision support system must be able to meet their various needs and desires, as well as provide various plans and strategies [5]. A study by [6] found that a smart tourism experience can improve satisfaction, so tourists want to look back at attractions. Mobile and smartphone technology has become the technology platform for the next generation of e-tourism recommendation applications [7]. Digital technologies can enrich the traveler experience across the travel cycle, opening up new channels for more responsible and sustainable consumption behavior [8]. The technology used in the development of the tourism industry will be able to improve the production efficiency of the tourism industry and make the information management platform smarter [9]. Regarding the various studies that have been conducted, it can be concluded that the application of digital technology, especially that provides smart systems, has the potential to improve tourism industry services by increasing tourist experiences.

C. RECOMMENDER SYSTEM

The availability of a recommender system (RecSys) is one of the efforts to be able to assist tourists in obtaining information on tourism products that suit their needs. With the increasing number of data related to tourism, there is a great opportunity to improve RecSys in the tourism sector [10], [11]. Various techniques have been used in previous studies including Collaborative Filtering (CF), Demographic Filtering (DF), and Hybrid Technique (HB). These techniques strive to enhance user experiences by leveraging data from past user interactions, assisting in the efficient retrieval of product and service information so that users no longer need to search for all the information needed through conventional methods that require efforts that take longer amidst the various sources of information available in the big data era.

Along with the growth of data and digital technology used for tourism activities as well as intelligent technology support through various techniques in machine learning, there are many opportunities for further studies to improve the service quality in the tourism industry such as the implementation of RecSys. Quality improvement of the RecSys is necessary for providing better services for users. The improvements can be provided through the data quality used in RecSys to produce a better performance of the RecSys model. CF is one of the approaches used in RecSys that provides recommendations for various products or services to users, but with the fact that not all users have interacted with every available product, so RecSys will face various obstacles related to cold-start problems. The lack of available ratings affects the accuracy of neighbourhood-based recommendation methods, and this problem is common to most RecSys because users usually rate only a small percentage of the available items [12].

Improving RecSys performance can also be achieved through a hybrid approach by combining some techniques,

some of which have been proposed by [13] and [14] to solve different problems using different techniques. The opportunities for improvements can be conducted using a combination of some other techniques regarding the performance improvement of RecSys. Today, social media provides the latest updates and trends about tourism destinations. It highly influences the tourism sector and attracts a lot of study [15]. Another improvement can be made through the utilization of user opinions based on their past experiences in using tourist products. User opinions provide valuable insights and serve as critical inputs for RecSys, enabling new rating generation to reduce matrix sparsity and offer aspect-based recommendations. By providing recommended items together with the related aspects, the users get more comprehensive information for decision-making. In this case, the implementation of techniques in Natural Language Processing is very necessary as part of developing a RecSys. All types of improvements in RecSys become valuable attributes for the improvement of the tourism industry through better and personalized tourist experiences. Kajian literatur lebih lanjut untuk menjawab beberapa Research Question dalam pengembangan RecSys pada industri pariwisata disajikan di bagian II dan III, termasuk tahapan pengembangan the RecSys model through the Cross-Industry Standard Process for Data Mining (CRISP-DM) secara koheren.

Based on tourism industry reports and studies, tourism experiences studies, and tourism recommendation systems studies mentioned above, several problems have been identified as follows:

- a. Sparse matrix is a common problem in RecSys, but various user reviews on the website along with Google Maps have not been utilized to solve this sparsity.
- b. Many tourists get repetitive, irrelevant and excessive information through conventional approaches on several types of media such as websites, social networks, brochures, travel agents and search engines.
- c. No study for RecSys combines CF, DF, and Sentiment Analysis techniques along with Content-Boosted Collaborative Filtering to solve the cold-start problem using combined quantitative data and qualitative data.

Regarding the identified problems, several Research Questions (RQ) have been prepared to propose the RecSys model as follows:

- a. How to overcome the cold-start problem in RecSys by developing new ratings by utilizing existing quantitative ratings and user reviews, as well as developing user demographic profiles that are still limited?
- b. How to develop a hybrid RecSys for the tourism industry using User-Based Collaborative Filtering (UBCF), Demographic Filtering (DF), Aspect-Based Sentiment Analysis (ABSA), and Content-Boosted Collaborative Filtering (CBCF)?

This study proposes a novel approach to enhance RecSys performance, utilizing a hybrid technique and Content-Boosted Collaborative Filtering (CBCF). The proposed model uses a dataset that was produced through

data processing that used the historical transactions from the TripAdvisor website. This hybrid approach combines the User-Based Collaborative Filtering (UBCF) technique with Demographic Filtering (DF) as well as the use of Aspect-Based Sentiment Analysis (ABSA) which converts the polarity of sentiment from user reviews to ratings. The dense matrix regeneration is conducted to improve the performance and solve the cold-start problem through the hybrid technique. Measurements of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are conducted to evaluate the RecSys performance.

The proposed hybrid RecSys model is expected to be an academic and practice reference to solve the cold-start problem in RecSys accompanied by a RecSys model framework and steps to solve the cold-start problem. The proposed model is also expected to be an academic and practice reference in developing RecSys in various other industry sectors, including entertainment, finance, health, and education.

II. TOURISM RECOMMENDER SYSTEM

A recommender system (RecSys) is a type of Artificial Intelligence [16]. A RecSys is a technology that provides recommended items to a user through any approach [12], [17]. There are some fundamental techniques mostly used for RecSys development, such as context-aware techniques, collaborative filtering techniques, deep neural networks-based approaches, semantic, and content-based methods [7], [18]. Collaborative Filtering (CF) recommends items to users based on the interests of other users with similar preferences, measured by the similarity of their past preferences [12]. Today, various techniques are known to be implemented in RecSys, including Item-based CF, User-based CF, Content-based CF (CCF), DF, CBCF, as well as various hybrid techniques that combine several of these techniques.

Previous studies in RecSys were proposed through different objectives, models, and data used. A study by [19] applied a CF-based RecSys combined with an approximate constraint satisfaction (ACS) process for automating word-of-mouth (WOM) effects and provided personalized travel-planning services to tourists based on ACS using tourists' preference ratings on some destinations through one hundred undergraduate student opinions in Korea. Using larger datasets from diverse demographic backgrounds, combined with CF and other techniques, can better represent users' diverse preferences and mitigate the cold-start problem. A prioritized and personalized RecSys through an Item-based CF technique using the MovieLens dataset, and 51 synthetic users was proposed by [20]. The users were grouped into 7 clusters using K-means for distance measurement of all points of data from the middle of the groups by using Euclidean Distance. The use of original users from various demographic backgrounds with the support of a combination of Item-based CF along with other techniques will better represent users with real diverse preferences and avoid the cold-start problem. A personalized hybrid tourism RecSys for building a personalized

tourism RecSys that suggests the most convenient Points of Interest (PoIs) to a tourist in a particular destination through a hybrid RecSys using CF, DF and Content-based Filtering (CB) was proposed by [21]. This study used historical data from TripAdvisor through a WebHarvy crawler to provide the dataset. Users must choose the activity types to search for (i.e., destinations, restaurants, and hotels) before getting a recommendation, so the automation of RecSys will be more useful for users in getting the recommendation. User opinions on the website will be more valuable for sentiment analysis regarding the current issues of users based on their past experiences in using the products. A study by [22] utilized Deep Learning to classify and recommend tourism types based on users' personalities (introvert, neutral, extrovert) and their preferred tourism activities (natural resources, cultural experiences, tour facilities, events). This study provided a system architecture of the deep learning and adaptive RecSys using a correlation graph for pairing personality type with tourism type. Further study using another technique through tourism experiences from similar users will provide new insight into the recommender system.

Another study in the RecSys by [23] proposed an architecture of Big Data to provide applications of typical Cultural Heritage (CH). The study provided a sequence of PoI recommendations through a Big Data architecture design to help tourists in their cultural experience. Cultural data from the Italian Ministry of Cultural Heritage and Activities (MiBACT) repository was used in this recommender engine based on a context-aware and hybrid strategy combined with corresponding metadata through three phases: pre-filtering (location, popularity, user interest) using the Naive Bayes Classifier, ranking (interest, mood, popularity), and post-filtering (exclusion of cultural object that out of range and not interested). The corresponding metadata was composed of authors, title, genre, history etc. At the ranking phase, the popularity was produced from a cultural object (CO) that was most photographed, and the mood for each co was calculated as the weighted sum of sentiments from social networks produced by analyzing the title, type of cultural object and author using appropriate NLP libraries, and the user data log was used in calculating of interest score. It used small data sets for the studies and the number of users used in the NASA-TLX for evaluation. The content analysis from social networks rather than its title will have more impact on the quality of user mood, and using some techniques in Machine Learning will offer more personalized RecSys in the future.

The RecSys studied by [24] proposes a simpler and more convenient travel search process through the Swarm intelligent algorithm. Experimental evaluation for efficiency and performance demonstration was conducted for hybrid location based RecSys for mobility and travel planning regarding personalized lists generation of points of interest, especially regarding cultural heritage. POIs and routes of travel were generated for more specific target users. This study conducted an experiment using the TripAdvisor dataset to build

individual user profiles to predict locations of interest according to the personal and group activity information produced from location-based social networks. The experiment was carried out using check-in history, interests, preferences, needs, and user profiles (age, gender, and occupation). Recommending a wider range of items will accommodate more diverse tourist needs in the next study.

The performance of a RecSys largely depends on the quality of the matrix which will be input into the recommendation process. Using only a few ratings will result in biased recommendations, and some solutions to this problem are to fill in missing ratings through default values, predictions using a content-based approach, and also through active learning techniques [12]. Some studies related to efforts to improve matrix quality have been conducted through various techniques. Study by [13] tried to increase the spreading ability and increase the User-Item Matrix (UI-Matrix) accuracy in film recommendations by using selected item features by the user which has been proposed through Content-Based Filtering using preference vectors. This study has produced a better input matrix and obtained a higher prediction accuracy of around 15.5% according to the values of RMSE and MAE. Reducing the missing ratings to improve predictions and reduce the average absolute error in film recommendations has also been carried out by [14] through UBCF and Item-Based CF. The application of reducing missing ratings has been able to improve predictions and reduce the MAE by around 10% to 15%. The word-of-mouth (WOM) effects through the user reviews are valuable input for RecSys which can be new insights in decision-making [12], [25]. However, the previous studies have not utilized user reviews as a new opportunity to increase UI-Matrix density regarding the performance improvement of RecSys.

III. THE PROPOSED APPROACH

This study applies the development of the RecSys model through the Cross-Industry Standard Process for Data Mining (CRISP-DM) as presented in Fig. 1. It has been applied in previous studies for different purposes [26], [27]. CRISP-DM consists of six stages which are: business understanding, data understanding, data preparation, modeling, evaluation, and deployment [28].

A. BUSINESS UNDERSTANDING

In the first phase of CRISP-DM, understanding the project's goals and requirements is crucial from a business perspective in the tourism industry. This knowledge is then transformed into a definition of the data mining problem and an initial plan to achieve these goals. Exploring the tourism industry is important to get a comprehensive understanding from a business perspective, such as understanding the needs of tourists. Understanding the tourism business from the tourist's point of view regarding their needs can uncover important factors that can influence the success of the project. Understanding the goals of each tourist, especially in planning tourism activity

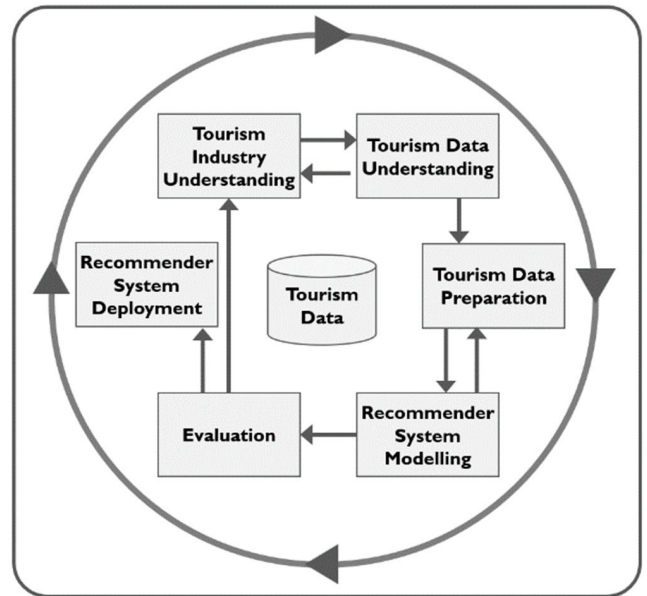


FIGURE 1. Cross-industry standard process for data mining (CRISP-DM) for tourism RecSys development.

through various media in obtaining tourist information will help in planning an appropriate project. Developing appropriate solutions in presenting more personalized information through a recommendation system is one of the solutions.

B. DATA UNDERSTANDING

In the second stage of CRISP-DM, data understanding starts with gathering data on tourism products and activities. It continues with tasks that enable understanding of the data, identification of problems related to data quality, discovery of initial insights, and identification of valuable subsets to make temporary assumptions about hidden information. A tourism recommender system development requires various data related to the tourism industry as well as data that technically meets the needs of the recommender system model. Various data sources can be involved, including the results of studies. The required data in both qualitative and quantitative formats consist of adequate information regarding feedback, tourism products and users. Preparation of initial data collection report is prepared at this stage with a description of the location, collection method, data relationship and the readiness of the data for the recommendation system model, and any problems that may arise.

C. DATA PREPARATION

The third stage of CRISP-DM includes all the tasks needed to build the final dataset for tourism RecSys that will be used for the modeling instrument from the initial raw data. The activities in this stage will usually be conducted multiple times and without a specific sequence. These activities are performed on tables, data rows, and the selection of attributes, along with the transformation and cleaning of data for

modeling instruments. This stage provides data preparation through data selection, data cleaning, data construction, and data format transformation which may be some or all the tasks to be performed. Data selection is carried out regarding attribute selection and record selection in a table. Data cleaning may involve selecting a clean subset of the data, embedding appropriate defaults, or more ambitious techniques such as estimating missing data with modeling. Constructive data may involve preparation operations such as the production of derived attributes or entire new records, or transformation of values for existing attributes. Data format transformations may involve syntactic modifications made to data that do not change its meaning but may be required by modeling tools.

D. MODELING

In the fourth stage of CRISP-DM, various modeling approaches are chosen and used, and parameter calibration is conducted to achieve the desirable values. Commonly, there are several approaches for the identical type of problem in data mining. Some approaches have special requirements for their data shape, so repeating this stage is often necessary. Regarding the RecSys development, previous studies have proposed various techniques. A study by [5] combined Knowledge-based filtering and Hybrid recommendation to develop a framework for a travel RecSys using literature reviews. The study found that tourists need an effective technological solution that meets their different requirements. Sparsity is one of the most important problems that can hurt the performance of recommenders [29] and the sparsity is a common problem with most recommender systems since users typically only rate a small portion of the available items [12]. By using 3 datasets, namely: Movielens, Netflix, and Epinions, the study applies k-NN to predict empty ratings by utilizing user similarities with other users to increase the reliability of the recommendation system. The performance of the rating prediction process with the MAE result of 0.604. Study by [30] using the Movielens and NetflixPrize datasets has shown that the RMSE value of the reference model recommendation with meta-MLP increases by 1%-10% when sparsity is reduced by 0.001%.

This study proposes a hybrid technique and CBCF as presented in Fig. 2, which consists of three steps of CBCF:

- CBCF-1: This CBCF aims to recommend a list of tourist attractions using the predicted UI-Matrix based on ratings from the original UI-Matrix.
- CBCF-2: This CBCF aims to recommend a list of tourist attractions using the predicted UI-Matrix based on ratings from the original UI-Matrix, combined with Demographic filtering (DF).
- CBCF-3: This CBCF aims to recommend a list of tourist attractions using the predicted UI-Matrix based on ratings from the original UI-Matrix, combined with Demographic filtering (DF) and Aspect-Based Sentiment Analysis (ABSA). ABSA is also used to explore

various aspects getting from user reviews so that it can complement the information provided by RecSys.

To get a performance comparison of the three CBCFs which are CBCF-1, CBCF-2, and CBCF-3, experiments were first conducted using the User-Based Collaborative Filtering (UBCF) technique which uses the original UI-Matrix. The original UI-Matrix is provided in the UBCF technique according to the results of the Data Preparation stage. The original UI-Matrix is used as an input for UBCF and followed by DF and ABSA which used the additional synthetic data generations combined with the original UI-Matrix. A hybrid technique together with CBCF is proposed for performance improvement of RecSys and solving the Cold-Start Problem. CBCF works by reducing the sparsity of the original UI-Matrix through three stages carried out serially, namely rating generation through k-NN prediction, rating generation through DF, and rating generation through rating generation from sentiment polarity in ABSA.

The UI-Matrix is provided from the dataset that has been produced in the previous data preparation stage which will become input in the recommendation process. The UI-Matrix describes the relationship between users and tourism products which contains user ratings of each product (item) in the CF model as represented in Table 1.

TABLE 1. User-item rating matrix for CF.

		List of Users			
		Item 1	Item 2	...	Item n
List of Users	User 1	R.1-1	R.1-2	..	R.1-n
	User 2	R.2-1	R.2-2	..	R.2-n
	...	...	...	..	...
	User m	R.m-1	R.m-2	..	R.m-n

User 1, User 2, and so on are a list of users who have already had transaction history, while Item 1, Item 2, and so on are a list of tourism products that are available and have been chosen by users. The user experience of tourism products is represented by each rating that connects users with tourism products including R.1-1 which contains the rating given by User 1 for tourism product 1 (Item 1), and R.1-2 which contains the rating given by User 1 towards tourism product 2 (Item 2), and so on. Based on the contents of the UI-Matrix, the recommendation process is carried out by measuring the level of similarity for each user. User similarity is an important stage in the CF technique for RecSys through the UI-Matrix.

This study obtained user similarity through the Pearson Correlation function. The Pearson Correlation function of user similarity is measured as follows:

$$PC(u, v) = \frac{\sum_{i \in I_{uv}} (r_{ui} - \bar{r}_u) (r_{vi} - \bar{r}_v)}{\sqrt{\sum_{i \in I_{uv}} (r_{ui} - \bar{r}_u)^2 \sum_{i \in I_{uv}} (r_{vi} - \bar{r}_v)^2}} \quad (1)$$

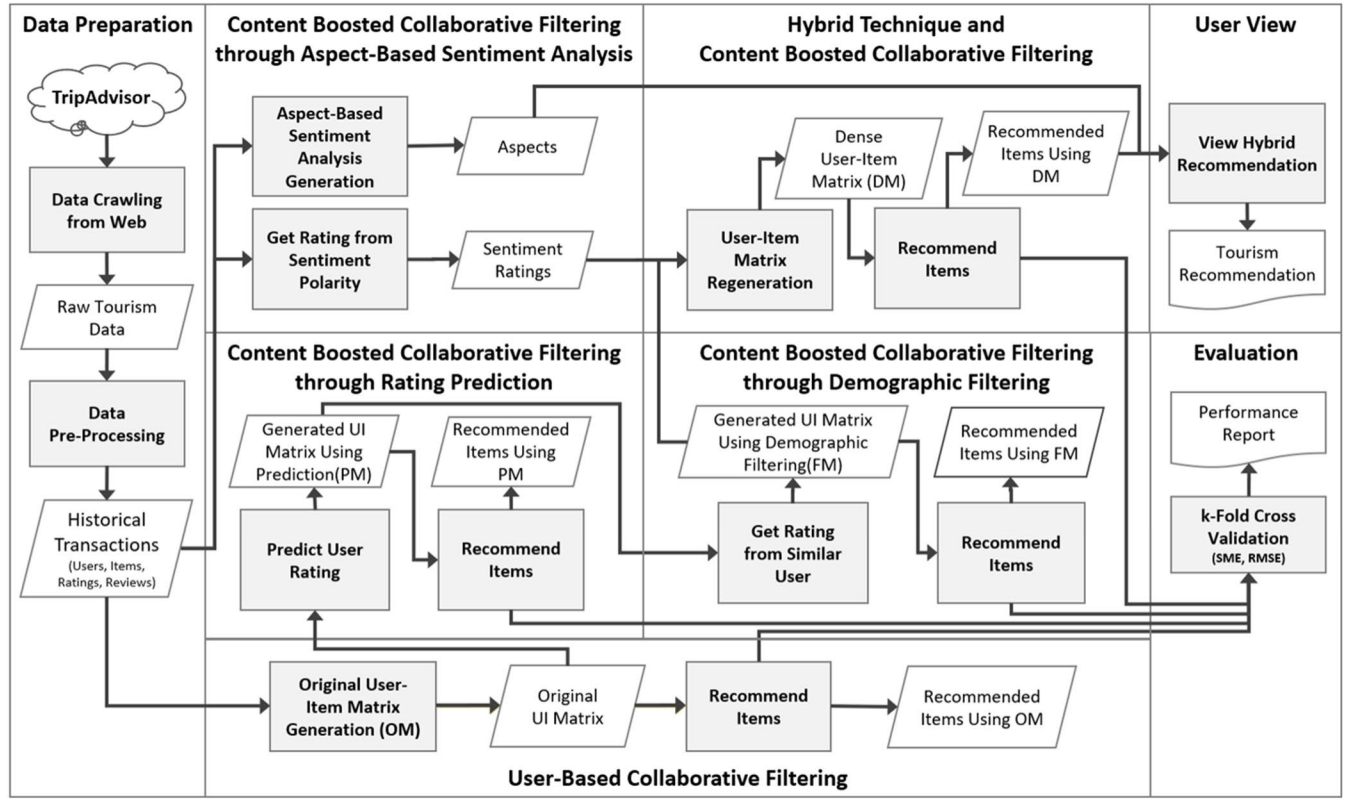


FIGURE 2. Cross-industry standard process for data mining (CRISP-DM) for tourism RecSys development.

where:

- $PC(u, v)$ is a similarity degree between user u with user v .
- r_{ui} is the rating given by user u for the item i .
- r_{vi} is the rating given by user v for the item i .
- $\bar{r}_u$ is the average rating of user u .
- $\bar{r}_v$ is the average rating of user v .
- $\mathcal{I}_{uv}$ is the item rated by user u and user v .

Prediction is obtained through the k-NN function:

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_{v \in \mathcal{N}_i(u)} w_{uv} (r_{vi} - \bar{r}_v)}{\sum_{v \in \mathcal{N}_i(u)} |w_{uv}|}, w_{uv} = \frac{1}{D(u, v)} \quad (2)$$

where:

- $\hat{r}_{ui}$ is the rating prediction for user u for the item i .
- $\bar{r}_u$ is the average rating of user u .
- r_{ui} is the rating for user u for the item i .
- $\mathcal{N}_i(u)$ is the set of neighbors of user u .
- $\mathcal{W}_{uv}$ is the similarity weight of user u with user v .
- $D(u, v)$ is the Euclidean distance between vector representation of user- u with user v .

The Pearson correlation is conducted to measure similarities between items based on given ratings by users in the past for tourism products or items. This measurement is conducted as the idea inspired by Ricci [12] to obtain the similarities between two items. Furthermore, the similarity

weight denoted by w_{uv} is conducted to permit trusted neighbor selection and provide the means to give importance to these neighbors in the prediction [12]. The ratings of the items given by each user are provided on a scale of 1 to 5 which represents the level of user satisfaction ranging from very unsatisfied, unsatisfied, quite satisfied, satisfied, and very satisfied. In the following process, this study conducts a prediction process using the k-NN algorithm which computes the distance between users represented as a vector through the Euclidean distance. The Euclidean distance denoted by D , from user u to user v represents the length of a line segment between the two users in the Euclidean space.

At the synthetic data development stage for UI-Matrix, first, the model checks the empty (sparse) rating whether exists in the current UI-matrix, the model will predict the rating using k-NN if the sparse matrix exists. Based on the predicted UI-Matrix, User Demographic Filtering (DF) is conducted if a sparse matrix exists in the predicted UI-Matrix. Finally, if the sparse matrix exists in the Filtered UI-Matrix, a new rating generation according to the polarity of sentiment is conducted using the Filtered UI-Matrix combined with user reviews from the historical data through ABSA.

ABSA was conducted to produce related aspects for every item using user reviews through lemmatization, oversampling, tokenization, and padding aspects. POS tagging was used in Wordnet Lemmatizer to obtain classified words into nouns, adjectives, verbs, and adverbs. An oversampling

technique was conducted to manage data balance and enough data. The Natural Language Toolkit (NLTK) library in Python was applied in conducting the tokenization of the dataset. To avoid problems that may occur when producing different lengths of sentences, the padding process was applied. This study uses the DistilBERT language model which is an approximation method of BERT (Bidirectional Encoder Representations from Transformers) which only uses 60% of the parameters of the BERT model but has better computing speed compared to BERT [31].

Using a limited number of ratings will result in biased recommendations, and some solutions that can be conducted to this problem are to fill in missing ratings through various techniques [12]. The previous study [8] proposed a solution to solve the MovieLens dataset sparsity by extracting category rates from the user-item matrix according to the user preferences and organizing these into vectors. The extraction was conducted through the calculation of the percentage of genres that have been evaluated by users.

This study proposes a new approach to reduce the sparsity of the original matrix regarding the performance improvement of the recommendation system and solving the cold start problem. Three steps are carried out sequentially as presented in Fig. 3.

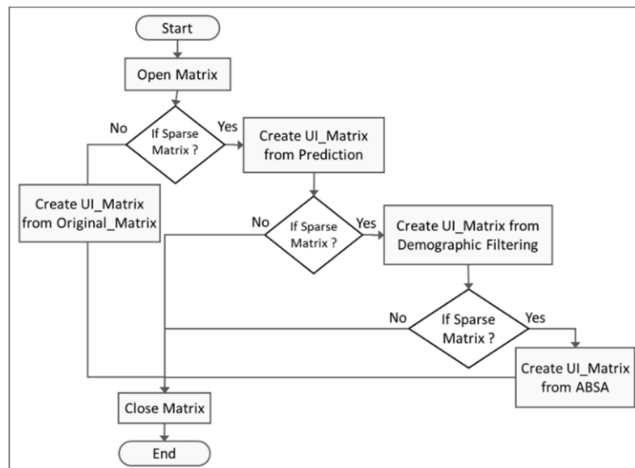


FIGURE 3. Cross-industry standard process for data mining (CRISP-DM) for tourism RecSys development.

In the first step, the model will check the sparsity of the original matrix, if a sparse matrix exists then the model will predict the new rating using k-NN based on the original UI-Matrix to produce the dense matrix. In the second step, the remaining sparse matrix that may exist from the first step is evaluated to conduct further matrix regeneration through demographic filtering (DF) to improve the fill-out of matrix cells. These two personalized approaches are conducted using the historical transactions of every user combined with other similar users. In avoiding the cold start problem and improving the density of the matrix cell produced by the second step, the following step is conducted through the rating generation from sentiment polarity produced by ABSA.

The first technique as presented in Fig. 4 uses the original UI-Matrix as input parameter for the algorithm. Using prediction and the original UI-Matrix this algorithm produces a new UI-Matrix through the output parameter with denser than the input matrix.

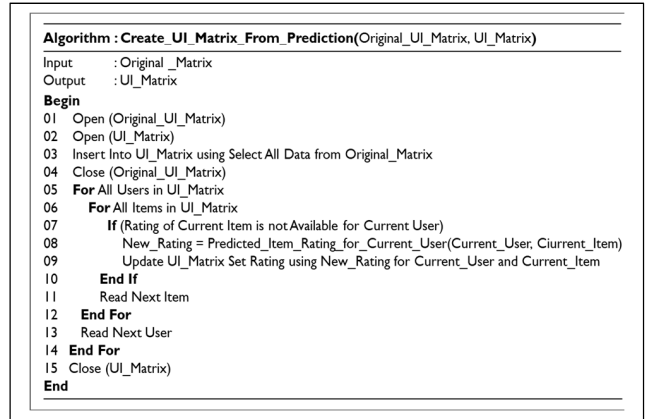


FIGURE 4. The algorithm for creating UI-Matrix using prediction.

The second technique as presented in Fig. 5 uses the UI-Matrix produced by the first one as input parameter for the algorithm. Using demographic filtering and the input matrix this algorithm produces a new UI-Matrix through the output parameter with denser than the input matrix. This algorithm uses the UI-Matrix as the input and output matrix.

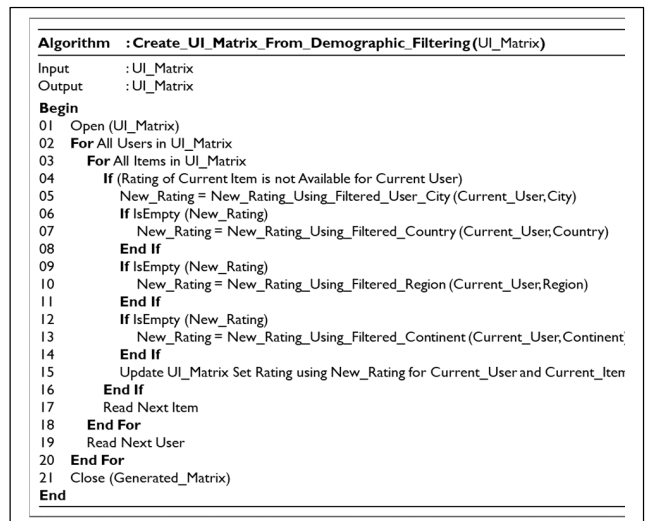


FIGURE 5. The algorithm for creating UI-Matrix using DF.

The third technique as presented in Fig. 6 uses the UI-Matrix produced by the second one as input parameter for the algorithm. Using ABSA and the input matrix this algorithm produces a new UI-Matrix through the output parameter with denser than the input matrix. This algorithm uses the UI-Matrix as the input and output matrix.

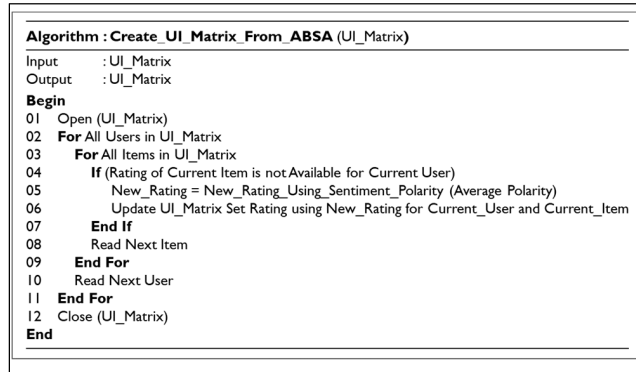


FIGURE 6. The algorithm for creating UI-Matrik using ABSA.

E. EVALUATION

In the evaluation phase, a high-quality model is developed from a data analysis perspective. Before conducting the finishing implementation of the model, it is necessary to evaluate it thoroughly and review the steps taken, to ensure it properly achieves the objectives of the business. The main objective is to ascertain whether some essential business issues have not been adequately examined. The proposed tourism RecSys model is evaluated by measuring the level of accuracy through the measurement of error level using MAE and RMSE parameters using K-Fold Cross Validation (K-Fold CF).

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (3)$$

where:

- n is the number of samples.
- Y_i is the mean actual rating of the i -th sample.
- $\hat{Y}_i$ is the predicted rating of the i -th sample.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (4)$$

where:

- n is the number of samples.
- Y_i is the mean actual rating of the i -th sample.
- $\hat{Y}_i$ is the predicted rating of the i -th sample.

F. DEPLOYMENT

In the final stage of implementation, decisions regarding the use of data mining results must be made. To enable customers to use the knowledge gained, organizing, and presenting the knowledge is needed at the implementation stage. This stage can be as easy as providing a report or as complicated as performing a repetitive data mining process across the organization. It is important for customers to first comprehend what activities need to be conducted to truly take advantage of the model provided. The deployment of the results of this study is conducted through the publication of datasets to Mendeley Data which can be accessed by researchers and managers regarding further research and insight for service improvement in the tourism industry.

IV. EXPERIMENTAL RESULT

This section describes the experimental research of tourism RedSys and some related scenarios. The experiment was conducted through a personal computer with specifications: RAM 8GB, 64-bit Operating System, x64-Based Processor.

A. BUSINESS UNDERSTANDING

The prospects and challenges of the tourism industry as explained in the introductory section need attention, especially regarding improving tourism products and services. The tourism industry has made a positive contribution to the regional economy even though it has faced very hard challenges during the COVID-19 pandemic. Service improvement in the industry sector is needed, including providing service for tourists regarding their travel planning. At this planning stage, the relevant information is needed in the flood of information available in various formats from various sources. Online social networks now play an important role in our daily lives and our decision-making and behaviour in many areas [32]. Online media with global reputations, such as the TripAdvisor website has been selected to obtain tourism information as well as being used to provide reviews of tourism products. The availability of tourism RecSys as a smart solution to provide personalized tourism products for every tourist is urgently needed instead of using conventional which needs hard effort for every user [33]. This study proposes a smart tourism RecSys to users in getting specific and more personalized products easily.

B. DATA UNDERSTANDING

Various sources of information have been chosen by tourists in planning tourism activities, from travel agents, brochures, web search engines, social networking media, and web-based applications. The TripAdvisor website provides tourism information together with user reviews and ratings. User opinions are very important in providing insights and knowledge that can then be used as input for tourist recommendations [34]. Considering that the data format is still scattered on the web page, it is necessary to conduct the data mining process followed by data processing. This research involves collecting tourism data from the TripAdvisor webpage, encompassing information on products, users, ratings, and user reviews. Data description was provided for data understanding through a list of entities, structure, format, domain values, and opportunities to learn in the RecSys. This information was collected through a WebHarvy module which can mine data from the TripAdvisor website and produce it in Excel format that consists of tourism products in Indonesia, specifically from the Malang, Yogyakarta, and Bali. This data contains tourism experiences from both domestic and foreign visitors. Next, the data was prepared through the initial processing according to the research objectives to support the development of the tourism RecSys model.

C. DATA PREPARATION

Based on the characteristics of the model that would be developed, data preparation has been conducted through the data sources both available on the TripAdvisor website and other data obtained from the search process on Google Maps for further explanation regarding user demographic information. The data preparation process was carried out through a crawling process, followed by data processing and validation. Data crawling from the TripAdvisor website yielded a single raw dataset available in Excel format, comprising tourist attractions, ratings, users, and reviews. Regarding fit with the model requirement, the data structure, format, and values from the crawled process needed further preparation and validation.

This data preparation and validation were conducted manually on raw data stored in Excel files by including additional sources that were also searched manually on Google Maps regarding descriptions of the user's location to obtain further explanation of the demographic aspects of the user. Several activities were carried out at this stage in which explaining the model, forming a matrix, deleting data that is not relevant to the model, filling in new data, encoding data values, and separating data into several special or unique attributes either into new attributes or new entities. The data preparation has produced a tourism dataset consisting of ten entities as the fundamental input in developing a RecSys which are: rating, user, item, type, review, mode, city, country, region, and continent. The dataset consists of 195,651 visit transactions and reviews, 88,194 users, 1,296 items, 96 attraction types, 5 visit modes, 9,142 cites, 164 countries, 21 regions, and 5 continents.

D. MODELING

The UBCF model was developed through the UI-Matrix which was produced from the dataset that consists of users' data, items, and ratings. This matrix represents the user-item interactions with a rating that a user has given to a certain item. The user-item interactions with the average rating that users have given to items in the past are presented in Table 2.

TABLE 2. UI-Matrix.

User ID	Items									
	Bali Zoo	Kuta Beach - Bali	Merapi Volcano	Sacred Monkey Forest Sanctuary	Sanur Beach	Seminyak Beach	Tanah Lot Temple	Tegalalang Rice Terrace	Uluwatu Temple	Waterbom Bali
568				3.5	5.0	3.0			5.0	4.0
1088		4.0		4.0		3.5	4.0	3.0	5.0	5.0
1941		3.5		3.7	3.0		3.0	4.5		4.8
8357	1.0			4.0	1.0	4.0	4.0		5.0	4.6
8848				4.5			4.0	4.8	4.0	4.0

Since some users' ratings tended to be higher or lower than others, a rating normalization process was necessary, involving the extraction of each user's average rating. The normalized matrix is presented in Table 3.

TABLE 3. Normalized UI-Matrix.

User ID	Bali Zoo	Kuta Beach - Bali	Merapi Volcano	Sacred Monkey Forest Sanctuary	Sanur Beach	Seminyak Beach	Tanah Lot Temple	Tegalalang Rice Terrace	Uluwatu Temple	Waterbom Bali
568				-0,6	0,9	-1,1			0,9	-0,1
1088		-0,1		-0,1		-0,6	-0,1	-1,1	0,9	0,9
1941		-0,3		-0,1	-0,8		-0,8	0,8		1,1
8357	-2,4			0,6	-2,4	0,6	0,6		1,6	1,2
8848				0,3			-0,3	0,5	-0,3	-0,3

In a normalized matrix, smaller ratings than the average user rating get a negative value, greater ratings than the average user rating get a positive value, the equal ratings to the average user rating get a value of zero. The matrix normalization process applied mean-centering notation [7] using the following function:

$$h(r_{ui}) = (r_{ui} - \bar{r}_u) / s_r \quad (5)$$

where:

- r_{ui} is the original matrix.
- $\bar{r}_u$ is mean of r_{ui} .
- s_r is standard deviation or r_{ui} .
- $h(r_{ui})$ is the normalized matrix.

Measuring the level of similarity between users using a function (1) was conducted in the next step. This step has produced a level of similarity between users, and the example result of this similarity is presented in Table 4.

TABLE 4. User similarity matrix.

User ID	568	1088	1941	2438	4390	8357
568	1.00	0.88	-0.53		0.5	-0.33
1088	0.88	1.00	0.16	0.38	0.99	0.91
1941	-0.53	0.16	1.00	-0.71	-0.17	0.62
2438		0.38	-0.71	1.00		
4390	0.5	0.99	-0.17		1.00	-0.03
8357	-0.33	0.91	0.62		-0.03	1.00

Through the configuration for user similarity, a threshold of 0.8 is used, which means the user must have a coefficient for Pearson Correlation of at least a similarity level of 0.8.

1) USER-BASED COLLABORATIVE FILTERING

Based on the User Similarity Matrix this study has succeeded in providing a recommended list of items for every user as presented in Table 5 with the top 3 recommended items.

TABLE 5. List of some recommended items through UBCF.

User ID	Recommended Attractions	Attraction Score (Weighted)	Mean Actual Rating	Predicted Rating
1088	Uluwatu Temple	0.46	5.00	4.53
1088	Waterbom Bali	0.13	5.00	4.20
1088	Merapi Volcano	0.08	-	4.15
1941	Tegalalang Rice Terrace	0.75	4.50	4.50
1941	Waterbom Bali	0.61	4.83	4.36
1941	Bali Zoo	-0.07	-	3.68

The new rating can be predicted for both items that have never been visited by users and items that have been visited using the filled-out original User-Item Matrix of 38%.

2) CONTENT BOOSTED COLLABORATIVE FILTERING

This study has succeeded in conducting a CBCF to improve the performance of the RecSys and overcome the Cold-Start Problem through a prediction for CBCF-1, DF for CBCF-2, and ABSA for CBCF-3 as presented in the following sections.

3) CONTENT BOOSTED CF THROUGH PREDICTION

The first step of CBCF (CBCF-1) involved rating prediction using function (2) for k-Nearest Neighbors (k-NN). The empty ratings in the original matrix have been reduced and produced a 72% filled-out rating in the matrix. The results of the recommendations through CBCF-1 that used a 72% filled-out rating in the matrix produced the emergence of several new recommended items as presented in Table 6 with the top 3 recommended items.

TABLE 6. List of some recommended items through CBCF-1.

User ID	Recommended Attractions	Attraction Score (Weighted)	Mean Actual Rating	Predicted Rating
1088	Merapi Volcano	0.81	-	4.96
1088	Uluwatu Temple	0.81	5.00	4.96
1088	Waterbom Bali	0.81	5.00	4.96
1941	Waterbom Bali	0.98	4.80	4.83
1941	Tegalalang Rice Terrace	0.65	4.50	3.73
1941	Bali Zoo	0.15	-	4.00

CBCF-1 produced different results for recommended items compared with UBCF due to the production result of higher predicted item ratings than UBCF in this step.

4) CONTENT BOOSTED CF THROUGH PREDICTION AND DF

The second step of CBCF (CBCF-2) was conducted through rating prediction and Demographic Filtering (DF) of Users. The matrix has more density than the result of CBCF-1 through this approach which produced a 99% filled-out rating in the matrix. The results of the recommendations through CBCF-2 consist of the emergence of several new recommended items as presented in Table 7 with the top 3 recommended items.

TABLE 7. List of some recommended items through CBCF-2.

User ID	Recommended Attractions	Attraction Score (Weighted)	Mean Actual Rating	Predicted Rating
1088	Merapi Volcano	0.81	-	4.96
1088	Uluwatu Temple	0.81	5.00	4.96
1088	Waterbom Bali	0.81	5.00	4.96
1941	Waterbom Bali	0.98	4.80	4.83
1941	Tegalalang Rice Terrace	0.39	4.50	4.24
1941	Bali Zoo	0.14	-	3.99

CBCF-2 produced different results for recommended items compared with UBCF and CBCF-1 due to the new

production result of higher predicted item ratings than UBCF and CBCF-1 that appeared in this step.

5) CONTENT BOOSTED CF THROUGH PREDICTION, DF, AND ABSA

The third step of CBCF (CBCF-3) was conducted through rating generation using rating prediction, DF, and ABSA. The matrix has a little bit more density through this approach which produced a 100% filled-out rating in the matrix. The results of the recommendations through CBCF-3 consist of the emergence of new recommended items as presented in Table 8 with the top 3 recommended items.

TABLE 8. List of some recommended items through CBCF-3.

User ID	Recommended Attractions	Attraction Score (Weighted)	Mean Actual Rating	Predicted Rating
1088	Merapi Volcano	0.81	-	4.96
1088	Uluwatu Temple	0.81	5.00	4.96
1088	Waterbom Bali	0.81	5.00	4.96
1941	Waterbom Bali	0.98	4.80	4.83
1941	Tegalalang Rice Terrace	0.39	4.50	4.24
1941	Bali Zoo	0.14	-	3.99

CBCF-3 produced similar results for recommended items compared with CBCF-2 due to the use of just a little bit more density matrix for producing the recommended items. This study also succeeded in extracting various aspects for every attraction from user reviews as presented in Table 9.

TABLE 9. Some extracted aspects of attractions.

Aspects	Bromo Tengger Semeru National Park	Kuta Beach - Bali	Merapi Volcano	Sacred Monkey Forest Sanctuary	Seminyak Beach	Tanah Lot Temple
access	2.5	4.6	3.6	4.5	4.6	4.7
car	4.5	4.1	4.6	4.4	4.8	4.5
driver	4.6	3.8	4.6	4.6	4.4	4.6
garbage		2.0				
guide	4.7	4.8	4.4	4.5		4.7
hotel	4.7	4.1	4.7	4.5	4.2	4.6
landscape	5.0		5.0	5.0		5.0

The average ratings for aspects of tourism attractions are useful for users regarding additional information in recommendations and insight for the other stakeholders. These are valuable insights for the tourism industry as some aspects that need attention for improvements like the access aspect to Bromo Tengger Semeru National Park and Merapi Volcano, and the garbage aspect in Kuta Beach – Bali.

E. EVALUATION

The models were evaluated by measuring their performance using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Repetitive measurement for both parameters was conducted to get stable results through 5 cycles using 10-Fold Cross Validation.

1) USER-BASED COLLABORATIVE FILTERING (UBCF)

User-Based Collaborative Filtering (UBCF) technique has succeeded in generating both recommended items that have never been visited by users and items that have been visited. Performance evaluation of UBCF using 10-Fold Cross Validation for MAE and RMSE has been conducted through 5 cycles with relatively stable performance results in each round for both MAE and RMSE as presented in Fig. 7.

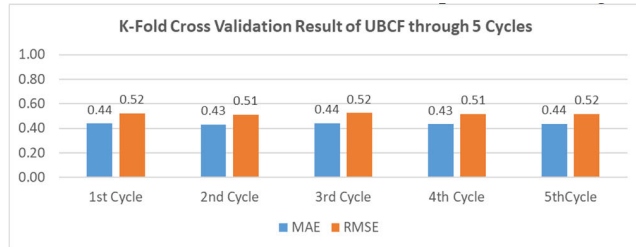


FIGURE 7. Performance evaluation of UBCF.

The RMSE result shows a higher level of sensitivity for each round compared to the MAE result. The average of UBCF performance for MAE and RMSE values respectively reach 0.4362 and 0.5180 using the 38% density of the original UI-Matrix.

2) CONTENT BOOSTED CF THROUGH PREDICTION

Using 72% denser of the UI-Matrix, the evaluation of the first CBCF (CBCF-1) was conducted using 10-Fold Cross Validation through 5 cycles with relatively stable performance results in each round for both MAE and RMSE as presented in Fig. 8.

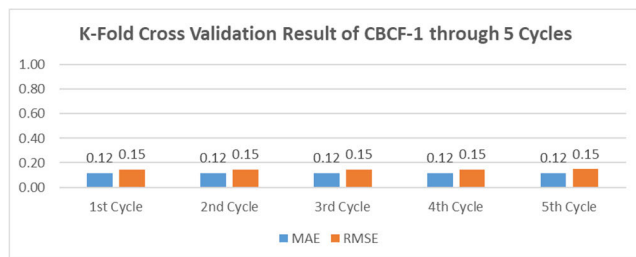


FIGURE 8. Performance evaluation of CBCF-1.

The CBCF-1 has increased the performance of the RecSys compared to UBCF using measurements of MAE and RMSE that produce the error decrement of around 73.6% for MAE of around 71.7% for RMSE. This performance improvement through the CBCF-1 is significant for RecSys to produce better tourism product recommendations for users.

3) CONTENT BOOSTED CF THROUGH PREDICTION AND DF

Using 99% denser of the UI-Matrix, the evaluation of CBCF-2 was conducted using 10-Fold Cross Validation through 5 cycles with relatively stable performance results in each round for both MAE and RMSE as presented in Fig. 9.

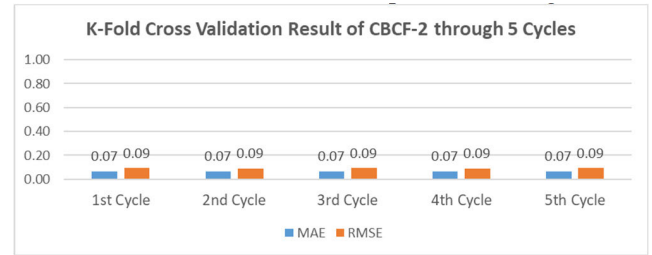


FIGURE 9. Performance evaluation of CBCF-2.

CBCF-2 has continued to increase the performance through prediction and DF by continuing to reduce the error level using the MAE with an error reduction of around 84.7% and continuing to reduce the error level using the RMSE of around 82.2% compared to achievement through UBCF. Compared with the achievements of CBCF-1, CBCF-2 has decreased the error level by around 11.1% for the MAE and has decreased the error level by around 10.5% for the RMSE. Based on the experiment results, the combination techniques in CBCF-2 have achieved better performance of the RecSys by reducing the error level compared to the UBCF and CBCF-1.

4) CONTENT BOOSTED CF THROUGH PREDICTION, DF AND ABSA

CBCF-3 has slightly improved the performance of RecSys using a slight increase in matrix density at 100% compared to CBCF-2. Using 100% denser of the UI-Matrix, the evaluation of CBCF-3 was conducted using 10-Fold Cross Validation through 5 cycles with relatively stable performance results in each round for both MAE and RMSE as presented in Fig. 10.

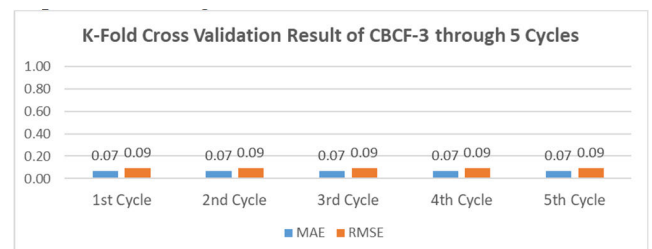


FIGURE 10. Performance evaluation of CBCF-3.

There are no significant result changes in CBCF-2 to CBCF-3, however, the CBCF-3 can avoid the Cold-Start Problem. Based on the achievements of MAE and RMSE, there is a slight decrease in error in CBCF-3 for MAE of around 84.7% compared to the UBCF result and for RMSE of around 82.3% compared to the UBCF result. Through CBCF-3, a list of recommended items has been produced which is obtained from other users' positive impressions of a tourist attraction based on their past reviews so that it can be ensured that each user will get a list of recommended tourist attractions. This study proposed a RecSys model using hybrid

technique and CBCF, and the performance summary of RecSys can be improved significantly compared with UBCF as presented in Fig. 11.

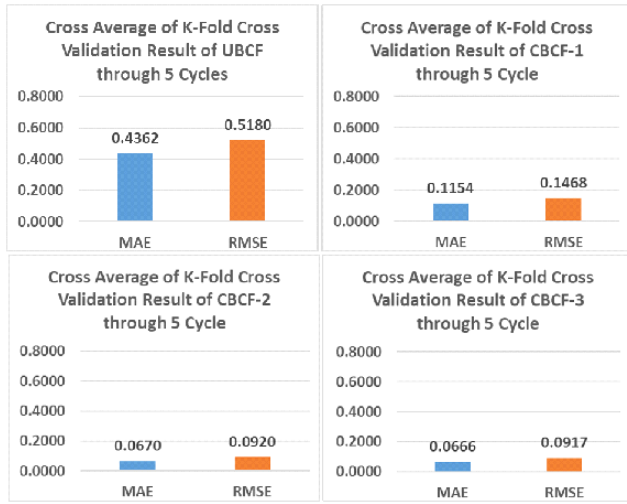


FIGURE 11. Performance summary result through 10-Fold CV.

Through the 5 cycles of the experiment, every technique has relatively stable results for the average of MAE and RMSE. UBCF has performance levels for MAE and RMSE at 0.4362 and 0.5180 using 38% density of original UI-Matrix, CBCF-1 at 0.1154 and 0.1468 using 72% density of UI-Matrix, CBCF-2 at 0.0670 and 0.0920 using 99% density of UI-Matrix, and CBCF-3 at 0.0666 and 0.0917 using 100% density of the UI-Matrix. Reducing the sparse matrix through the proposed model has been able to improve the performance of RecSys as presented in Fig. 12.

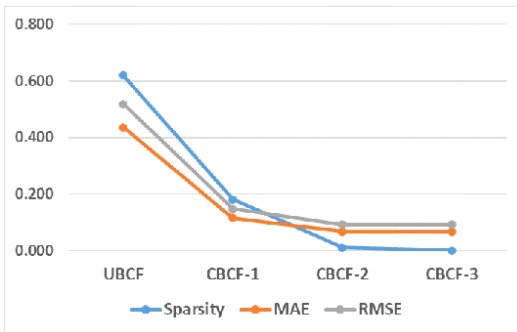


FIGURE 12. Performance results with different sparse UI-Matrix.

The proposed hybrid and CBCF models have been able to significantly improve the performance of the recommender system both through MAE and RMSE measurement results. This study has succeeded in proposing a novelty model for tourism recommender systems in avoiding the cold-start problem, sparse matrix reduction through the additional synthetic data generation for performance improvement complemented with various aspects of attractions in recommended items.

F. DEPLOYMENT

By utilizing raw data crawled from the TripAdvisor website for dataset development, this study successfully provided a dataset, parts of which have been published on Mendeley Data with the DOI 10.17632/h58s544674.1. The dataset can be used for further study in RecSys and become insights for managers of attractions regarding service improvement. The dataset consists of related information on users, items, demographics, transactions, and ratings. The entity relationship diagram of the dataset that has been produced through this research is presented in Fig. 13.

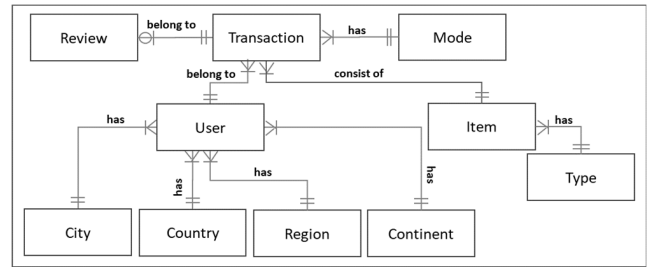


FIGURE 13. Entity relationship diagram for tourism dataset.

The entity relationship diagram consists of 10 entities which are: User, City, Country, Region, Continent, Transaction, Review, Mode, Item, and Type. The user entity represents the information of tourist consist of the following attributes: UserId, ContenentId, RegionId, CountryId, and CityId. City entity consists of the following attributes: CityId, CityName, and CountryId. The user entity that provides the information of tourists consists of the following attributes: UserId, ContinentId, RegionId, CountryId, and CityId. The city entity that provides the information city consists of the following attributes: CityId, CityName, and CountryId. The country entity that provides the information of country consists of the following attributes: CountryId, CountryName, and RegionId. The region entity that provides the information of region consists of the following attributes: RegionId, RegionName, and ContentId. The continent entity that provides the information of continent consists of the following attributes: ContenentId, ContenentName. The transaction entity that provides the information transaction consists of the following attributes: TransactionId, UserId, VisitYear, VisitMonth, VisitModeId, AttractionId, and Rating. The review entity that provides the information of user's review consists of the following attributes: TransactionId, AttractionId, AttractionCity, AttractionType, UserId, ReviewYear, ReviewMonth, and UserReview. The mode entity that provides the information of visiting mode consists of the following attributes: VisitModeId, VisitMode. The item entity that provides the information of attraction or item consists of the following attributes: AttractionId, AttractionCityId, AttractionTypeId, AttractionName, and AttractionAddress. The type entity that provides the information of attraction type consists of the following attributes: Attraction-TypeId, AttractionType.

V. CONCLUSION

This study successfully applied the CRISP-DM methodology to develop a tourism RecSys model. This hybrid model combines UBCF, DF, and ABSA with the CBCF technique. The proposed RecSys model offers personalized tourism product recommendations, circumvents the Cold-Start Problem, and enhances the RecSys performance by lowering error levels, as evidenced by MAE and RMSE results, in comparison to using UBCF alone. Based on the average level of MAE and RMSE through the 5 cycles of 10-Fold CV, the UBCF has respectively performances at 0.436 and 0.518 using the 38% density of the original UI-Matrix. Gradually, the performance of RecSys continues to increase, as shown by the continued decrease in error rates in CBCF-1, CBCF-2, and CBCF-3 as the sparsity in the UI-Matrix decreases. The following first developed model though the CBCF-1 has respectively performances at 0.115 and 0.147 using 72% density of UI-Matrix. The second developed model through the CBCF-2 has respective performances at 0.067 and 0.092 using the 99% density of UI-Matrix. The last developed model through the CBCF-3 has respective performances at 0.007 and 0.009 using 100% density of UI-Matrix. This performance improvement of RecSys using the denser UI-Matrix has a similarity with the result studies that have been conducted by [8] and [9].

This study produced a tourism dataset comprising activities in Bali, Malang, and Yogyakarta, Indonesia, enriched with additional geographic user information from Google Maps. The dataset has been published on the Mendeley Data that can be used for further studies in tourism RecSys through the various techniques. Various aspects related to tourism activities have been successfully mined from user reviews to become valuable insights for users in tourism product recommendations and related stakeholders. This study also has succeeded in proposing a novelty model for producing synthetic ratings as the additional ratings of the original UI-Matrix regarding the performance improvement of RecSys for other subjects.

Based on the achievement of this study, the performance of a RecSys can be improved through a combination of machine learning techniques and CBCF techniques by utilizing quantitative data and qualitative data from reviews that are transformed into quantitative data by utilizing two data sources, namely TripAdvisor and Google Maps to solve the cold-start problem. The hybrid technique through the proposed CBCF is a novelty that can improve the performance of RecSys with the decreasing level of errors produced along with the decreasing level of sparsity of the UI-Matrix

Theoretical implications that can be taken from the achievements of this study include the development of a tourism recommendation system model along with datasets generated from two formats, namely quantitative data and qualitative data taken from two different data sources. This study proposed practical implications through the stages in the development of a tourism recommendation system model

using a hybrid model that includes UBCF, DF, and ABSA techniques accompanied by steps in CBCF to provide recommendations for personalized tourism products according to past experiences and interests of each tourist and avoid cold-start problems. The proposed model can improve the performance of the RecSys as achieved from the results of MAE and RMSE. This study has also succeeded in providing an alternative to reduce the sparsity of the UI-Matrix which is very fundamental for the development of a RecSys that utilizes content according to tourist visit history data. The proposed RecSys model is a novelty that can be applied in the development of RecSys for various products or services in various other industrial sectors such as entertainment, finance, health, and education. The business implications for the tourism industry from the resulting model are expected to help the tourism industry, especially in improving services to tourists in obtaining recommendations for relevant tourism products from other tourists who have similar preferences and demographics. The proposed RecSys is also able to mine various aspects that are of concern to tourists and can be an important input for tourism destination managers to improve their services to tourists.

In summary, this study has succeeded in proposing a RecSys in producing a list of recommendations for user tourism products through a hybrid technique accompanied by CBCF. The reduction of sparsity in the UI-Matrix has been achieved through the proposed model and can improve the performance of RecSys. This study has succeeded in proposing a novelty in reducing sparsity in the UI-Matrix regarding the performance improvement of RecSys using quantitative data format and qualitative data format based on historical user ratings and user reviews. The proposed model can be applied to various other industry sectors such as entertainment, finance, health, and education.

The implementation of ABSA in this study uses the public opinion of tourism products to avoid the cold-start problem in general, so, further improvement is needed in the next study to provide more personalized recommendations. Providing application software that can be used directly by tourists to interact with the recommendation system along with providing features that can be used by tourists to continuously update their profiles and preferences will be able to improve RecSys so that efforts to solve the cold-start problem can be provided in a more personalized way. Further studies can also be conducted by including various additional user profiles including hobbies, professions, and visiting modes. Due to limited computing resources during the experiment, this study utilized a subset of the produced dataset. Regarding this limitation, it is necessary to provide computing technology resources that have higher specifications to be used in conducting experiments on all records of the dataset.

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